# EUROPEAN RESEARCH COUNCIL

Starting Grant

# Topological Quantum Error Correction via Machine-Learned Decoders

## TQML

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| **Grant Type** | StG |
| **Call Identifier** | ERC-2026-StG |
| **Duration** | 60 months |
| **Principal Investigator** | Dr. Elena Vasquez |
| **PhD Date** | March 2019 |
| **Host Institution** | ETH Zurich, Switzerland |
| **ERC Panel** | PE2 |
| **Requested Budget** | EUR 1,490,000 |
| **Additional Funding** | EUR 0 |

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## Abstract

Quantum computing stands at an inflection point. Hardware platforms based on superconducting qubits have demonstrated below-threshold error correction for the first time, yet the decoder technology required to sustain logical qubits in real time remains a critical bottleneck. Existing decoders — based on Minimum Weight Perfect Matching and Union-Find algorithms — were designed for idealized noise models and cannot efficiently handle the correlated errors, leakage, and crosstalk that characterize real quantum hardware. TQML (Topological Quantum Error Correction via Machine-Learned Decoders) proposes a paradigm shift: training deep reinforcement learning agents to discover optimal decoding policies for topological codes that adapt to the true, device-specific noise of current quantum processors. TQML pursues four objectives: (O1) developing a novel transformer/graph neural network decoder architecture meeting the one-microsecond real-time latency constraint via FPGA implementation; (O2) demonstrating that ML-decoders achieve strictly lower logical error rates than MWPM on realistic hardware noise including correlations and leakage; (O3) establishing a rigorous PAC-learning theoretical framework connecting neural decoder representational capacity to the information-theoretic limits of quantum error correction — founding a new subfield at the intersection of learning theory and quantum information science; and (O4) deploying and validating ML-decoders in closed-loop operation on a live superconducting quantum processor. If successful, TQML will remove one of the last major obstacles to fault-tolerant quantum computing, open a new theoretical domain, and provide the community with open-source tools that will accelerate decoder development globally. Dr. Elena Vasquez, independent group leader at ETH Zurich with publications in Nature Physics and Science and three consecutive invited talks at QIP, is uniquely positioned to lead this project.

Keywords: quantum error correction, topological codes, surface codes, machine learning, neural decoders, reinforcement learning, fault-tolerant quantum computing, FPGA, PAC learning, superconducting qubits

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# PART B1 — Part I of the Scientific Proposal

## Vision, Concept and Objectives

The central ambition of TQML is to resolve one of the most consequential open problems at the frontier of quantum information science: whether machine learning can fundamentally transform quantum error correction by discovering optimal decoding strategies for topological codes that surpass all known analytical approaches in both accuracy and real-time speed. Quantum computing is undergoing a transition from proof-of-principle demonstrations to devices of practical relevance. Yet a single critical bottleneck threatens to stall this transition: the inability to perform error correction fast enough, and accurately enough, to sustain logical qubits over computationally meaningful timescales. TQML addresses this bottleneck by proposing a paradigm shift — replacing handcrafted decoding algorithms with deep reinforcement learning agents that discover decoding policies autonomously, adapting to the true complexity of device-level noise.

The state of the art in quantum error correction decoding rests on two pillars: Minimum Weight Perfect Matching (MWPM) [6] and Union-Find algorithms. Both were designed for idealized, independent noise models and achieve their performance guarantees only under assumptions that real quantum hardware systematically violates. Modern superconducting processors — including Google's Willow [8] and IBM's Heron — exhibit correlated errors, leakage to non-computational states, and crosstalk between neighboring qubits. These phenomena invalidate the independence assumptions underlying MWPM and Union-Find, leading to logical error rates that are substantially higher than theoretical thresholds predict [4,5]. No existing analytical decoder can efficiently handle the full complexity of realistic, device-specific noise while simultaneously meeting the latency requirement of approximately one microsecond imposed by the coherence time hierarchy of superconducting qubits.

TQML proposes that this impasse is precisely where machine learning offers a qualitatively new solution. Rather than designing a decoder from first principles, TQML will train neural decoder agents directly on syndrome data generated by high-fidelity noise models extracted from current quantum hardware. The key scientific conjecture driving the project is that the representational capacity of deep neural networks — specifically transformer architectures augmented with graph neural network components matched to the lattice geometry of topological codes — is sufficient to implicitly learn the structure of correlated noise and exploit it for correction, in a way that surpasses any polynomial-time classical decoder operating under independence assumptions. Preliminary evidence from the PI's own work [1,2] supports this conjecture for small code distances; TQML will extend it to practically relevant distances and establish its theoretical foundations through a novel PAC-learning framework [9].

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The four main objectives of TQML are: (O1) Develop a novel neural network architecture optimized for topological code decoding that operates within the real-time latency constraint of approximately one microsecond when implemented in FPGA firmware. (O2) Demonstrate empirically that ML-decoders achieve strictly lower logical error rates than MWPM on realistic noise models extracted from current superconducting quantum hardware, including correlated and leakage noise channels. (O3) Establish a rigorous theoretical framework connecting the representational capacity of neural decoder classes to the information-theoretic limits of quantum error correction, yielding PAC-learning bounds for decoder families over surface codes and color codes. (O4) Deploy and validate ML-decoders in closed-loop operation on a live superconducting quantum processor in partnership with a major hardware provider, demonstrating below-threshold logical error rates using learned decoding policies.

These objectives are ambitious by design: O1 targets a latency regime that current neural network inference pipelines do not achieve without architectural innovation; O2 makes a falsifiable comparative claim against the established state of the art; O3 opens an entirely new theoretical subfield at the intersection of PAC-learning theory and quantum information science; and O4 constitutes a hardware demonstration that would represent a landmark result in the field. Taken together, they define a research agenda that could not be incrementally extracted from any existing programme.

## Groundbreaking Nature and Potential Impact

TQML is groundbreaking along three distinct and mutually reinforcing axes.

First, it redefines the algorithmic foundations of quantum error correction. The field has operated under the assumption that decoders must be analytically designed to exploit the known structure of stabilizer codes [4,5,7]. TQML challenges this assumption by positing that the correct structure to exploit is not the code geometry alone, but the joint geometry of code and noise — a structure that is device-specific, time-varying, and analytically intractable, but learnable. If this conjecture is confirmed, it constitutes a conceptual break from five decades of coding-theory-driven decoder design. Existing neural decoder work [10] has been limited to small code distances and idealized noise; TQML is the first programme to target practically relevant distances under realistic correlated noise, with a clear path to hardware deployment.

Second, TQML establishes a new theoretical territory. The question of which quantum error-correcting code families admit efficiently learnable decoders has never been posed in the language of computational learning theory [9]. The PAC-learning framework that TQML will develop for decoder classes is, to the PI's knowledge, the first systematic attempt to bring sample complexity and generalization theory to bear on quantum error correction. This is not an incremental theoretical refinement — it is the founding of a new subfield with its own open problems, proof techniques, and connections to quantum Shannon theory.

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Third, TQML targets a demonstration that the field has explicitly identified as a critical missing capability: real-time, hardware-adaptive decoding on a live quantum processor. The gap between the best academic decoder results and what is deployable in the control stack of a real device is large and has resisted closure precisely because it requires simultaneous mastery of machine learning, quantum hardware physics, and digital systems engineering. The TQML team's unique combination of expertise — evidenced by the PI's prior publications spanning neural decoders [1,2] and superconducting noise characterization [3] — positions this project to close that gap for the first time. The timing is decisive: quantum hardware has crossed the threshold at which error correction is no longer a theoretical exercise but an experimental necessity [8], making 2026–2031 the window in which decoder technology will determine whether fault-tolerant quantum computing is achievable on a practical timeline.

## Strategy

The overall research strategy of TQML unfolds in four overlapping phases, each building on the results of the previous while maintaining sufficient independence to allow continued progress if any single phase encounters obstacles. The foundational strategy is to couple empirical and theoretical research tightly from the outset, rather than treating theory as a post-hoc rationalization of experimental results.

In Phase 1 (M1–18), the project focuses on architectural innovation (WP1): designing transformer-based decoder networks with graph neural network components whose inductive biases are matched to the two-dimensional lattice structure of surface and color codes. Training will be conducted on simulated syndrome data generated by realistic noise models, including depolarizing noise, amplitude damping, correlated two-qubit error channels, and leakage. The key performance target is sub-microsecond inference latency, which requires innovations in model compression, quantization, and hardware-aware architecture search. In Phase 2 (M12–36), the project transitions from simulated to real hardware noise (WP2). Through IBM Quantum's publicly accessible device calibration data — providing an immediate starting point — and a direct research partnership with IBM Quantum or Google Quantum AI, the project will extract empirical noise process matrices from current processors and use them to retrain and fine-tune the learned decoders. This phase produces the direct comparison between ML-decoders and MWPM on hardware-realistic noise, which constitutes the central empirical claim of the project (O2).

Phase 3 (M24–48) develops the theoretical framework in parallel (WP3). The PI will lead a programme establishing PAC-learning bounds for quantum decoder classes, characterizing the sample complexity required to learn a near-optimal decoding policy for a given code family and noise model. This theoretical work is designed to be self-contained and to produce publishable results regardless of the hardware outcomes of Phase 2, ensuring scientific productivity across the project. Phase 4 (M36–60) implements the best-performing

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decoders in FPGA firmware and demonstrates closed-loop error correction on a live quantum processor (WP4). This phase is the integration of all preceding work and constitutes the landmark demonstration that the project is designed to achieve. The strategy ensures that even if FPGA latency targets require a hybrid NN+MWPM fallback architecture, the scientific contributions of the preceding three phases remain fully intact and publishable. Dr. Elena Vasquez will dedicate a minimum of 50% of her working time to TQML throughout its duration, in accordance with ERC Starting Grant requirements.

## References


[1] Vasquez, E. et al. (2021). Neural network decoders for small-distance surface codes. Nature Physics, 17(4), 429–437.

[2] Vasquez, E. et al. (2023). Adaptive decoding strategies for topological codes under correlated noise. Nature Physics, 19(8), 1142–1151.

[3] Vasquez, E. et al. (2022). Noise characterization and process tomography in superconducting transmon circuits. Science, 378(6621), 743–748.

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[9] Valiant, L. G. (1984). A theory of the learnable. Communications of the ACM, 27(11), 1134–1142.

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# Curriculum Vitae and Track Record

## Education and Key Qualifications

Dr. Elena Vasquez completed her PhD in Quantum Information and Condensed Matter Physics in March 2019. Her doctoral research focused on the theoretical and numerical analysis of stabilizer codes under realistic noise models, with a secondary focus on classical simulation methods for noisy quantum circuits. Prior to her doctorate, she obtained an MSc in Theoretical Physics and a BSc in Physics and Mathematics, both with distinction. Her interdisciplinary training spanning quantum information theory, condensed matter physics, and machine learning methods provides the precise combination of competencies required to execute the TQML research agenda.

## Current and Previous Positions

Following her PhD, Dr. Vasquez was awarded the ETH Zurich Postdoctoral Fellowship (2020), one of the most competitive internal fellowships at ETH, which supported two years of independent research at the Institute for Theoretical Physics. She subsequently transitioned to an independent research group leader position at ETH Zurich, where she currently leads a group of two postdoctoral researchers and three PhD students. This position is fully independent from her doctoral supervisor, who is based at a different institution and works in a distinct research direction (condensed matter theory). The transition to an independent group leader role in 2022 marks the clear origin of her autonomous research trajectory, and all publications from 2020 onwards originate from her independent research. She held the Google PhD Fellowship (2017–2019) during the final stage of her doctorate, demonstrating early international recognition of her scientific potential.

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## Research Achievements

1. Vasquez, E. et al. (2023). Adaptive decoding strategies for topological codes under correlated noise. Nature Physics, 19(8), 1142–1151. [Lead and corresponding author; introduced the first neural decoder demonstrating above-threshold performance on correlated noise models extracted from real superconducting devices.]

2. Vasquez, E. et al. (2021). Neural network decoders for small-distance surface codes. Nature Physics, 17(4), 429–437. [Lead and corresponding author; first systematic comparison of neural and analytical decoders for distance-3 and distance-5 surface codes.]

3. Vasquez, E. et al. (2022). Noise characterization and process tomography in superconducting transmon circuits. Science, 378(6621), 743–748. [Lead and corresponding author; developed a scalable protocol for extracting full process matrices of multi-qubit superconducting devices.]

4. DecoderBench: Open-source benchmark suite for quantum error correction decoders. GitHub repository with 500+ stars, adopted as a community standard for decoder performance evaluation.

[Primary developer; provides standardized noise models, syndrome generators, and logical error rate estimators used by groups worldwide.]

5. Invited talk: Quantum Information Processing (QIP) 2024. Talk title: 'Toward Real-Time Neural Decoding: Architecture, Latency, and Hardware Constraints.' [Invited by programme committee; QIP is the premier international conference in quantum information.]

6. Invited talk: Quantum Information Processing (QIP) 2023. Talk title: 'Learning Optimal Decoding Policies Under Correlated Noise.' [Invited; presented results leading to the 2023 Nature Physics publication.]

7. Invited talk: Quantum Information Processing (QIP) 2022. Talk title: 'Neural Network Decoders Beyond the Phenomenological Noise Model.' [Invited; three consecutive years of invited QIP presentations is an exceptional indicator of field recognition.]

8. Google PhD Fellowship in Quantum Computing (2017). [Awarded competitively to outstanding PhD students globally; one of approximately 15 fellowships awarded in the physical sciences in 2017.]

9. ETH Zurich Postdoctoral Fellowship (2020). [Awarded on the basis of a competitive research proposal; one of the most selective internal fellowships at ETH Zurich, supporting fully independent postdoctoral research.]

10. PhD thesis distinction and department prize for outstanding doctoral dissertation (2019). [Awarded by the Department of Physics for the most distinguished doctoral thesis of the academic year.]

## Additional Information

Dr. Vasquez's independence from her doctoral supervisor is unambiguous: she moved to a different institution (ETH Zurich) immediately upon completing her PhD, her research direction diverged substantially from her supervisor's group (which focuses on condensed matter theory rather than quantum error correction and machine learning), and all publications listed above were produced under her independent leadership at ETH. Her current group of five junior researchers (two postdocs, three PhDs) demonstrates established mentorship and leadership capacity. She has co-organized a workshop on machine learning methods in quantum error correction at the Erwin Schrödinger Institute (Vienna, 2023), further evidencing her role as a recognised leader in this emerging area. The open-source DecoderBench suite — with over 500 GitHub stars and adoption by research groups at MIT, Caltech, and Delft — demonstrates her commitment to open science and her ability to produce community-level impact beyond individual publications.

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# PART B2 — Part II of the Scientific Proposal

## Research Methodology

The TQML methodology integrates four interlocking research streams: neural architecture design, hardware noise extraction and validation, theoretical foundations of learnable decoding, and real-time FPGA deployment. Each stream produces independently publishable outputs while contributing to the integrated demonstration in WP4.

Stream 1 — Neural decoder architecture (WP1). The core architectural hypothesis is that a hybrid transformer/graph neural network (GNN) architecture, where the GNN encodes the lattice geometry of the surface code and the transformer encodes temporal correlations across syndrome measurement rounds, will outperform both purely sequential and purely spatial architectures. The GNN component will use the stabilizer check graph as its inductive bias, with node features representing syndrome bits and edge features representing potential error paths weighted by prior noise probabilities. The transformer component will process sequences of syndrome snapshots across multiple measurement rounds, capturing the temporal correlations that arise from leakage and other non-Markovian noise sources. Training will use a combination of supervised learning on labeled syndrome-to-correction mappings and reinforcement learning with logical error rate as the reward signal. The RL phase is essential because the supervised objective (correctness of individual Pauli corrections) is only a proxy for the true objective (minimization of the logical error rate), and the two objectives can diverge substantially under correlated noise. Architecture search will be guided by a latency-accuracy Pareto frontier, with pruning, quantization, and hardware-aware neural architecture search applied iteratively to push the inference time toward the target of one microsecond.

Stream 2 — Hardware noise modeling and validation (WP2). Real noise profiles will be extracted from superconducting quantum processors using two complementary strategies. First, IBM Quantum's publicly available device calibration data — including single-qubit and two-qubit gate fidelities, T1 and T2 times, and readout error rates — will be used to construct device-specific noise models for immediate use from M12. Second, through a direct research partnership with IBM Quantum or Google Quantum AI, the project will access raw syndrome data from repeated stabilizer measurements on current devices, enabling extraction of correlated error probabilities beyond what is captured by standard process tomography [3]. The decoder trained on simulated noise will be fine-tuned on this hardware syndrome data using transfer learning techniques, testing the hypothesis that a decoder pre-trained on high-fidelity simulations can rapidly adapt to real device noise with limited hardware data.

Stream 3 — Theoretical foundations (WP3). The theoretical programme will develop a PAC-learning framework [9] for quantum decoder classes. The central question is: given a noise model drawn from some distribution over noise channels consistent with a physical device

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model, how many syndrome samples are required to learn a decoding policy that achieves logical error rate within epsilon of the optimal decoder, with probability at least 1 − delta? This requires defining appropriate hypothesis classes of decoder functions (parameterized neural networks of bounded complexity), establishing VC-dimension or Rademacher complexity bounds for these classes, and connecting these classical learning-theoretic quantities to the quantum channel capacity of the code. A secondary theoretical direction will investigate the computational complexity of optimal decoding under correlated noise models, with the goal of establishing hardness results that formally justify the use of heuristic ML approaches [7].

Stream 4 — FPGA deployment (WP4). The FPGA implementation will target a commercial FPGA platform (Xilinx Ultrascale+ or equivalent) integrated into the classical control stack of a superconducting quantum processor. The neural network weights from the best-performing architecture identified in WP1 will be implemented using fixed-point arithmetic with precision determined by the accuracy-latency trade-off analysis. Pipelined inference will be implemented to maximize throughput. The target latency of approximately one microsecond for a distance-5 surface code decoder is aggressive but physically motivated: it corresponds to one-tenth of the typical T1 time of current transmon qubits, leaving sufficient margin for ancilla reset and gate operations within a single error correction cycle.

## Work Plan

| WP | Title | Period | Summary |
|---|---|---|---|
| WP1 | Neural Decoder Architecture and Training Pipeline | M1–M24 | Design, implement, and optimize transformer/GNN hybrid decoder architectures for surface and color codes. |
| WP2 | Hardware Noise Modeling and Experimental Validation | M12–M36 | Extract device-specific noise profiles from IBM Quantum and partner hardware. |
| WP3 | Theoretical Foundations of Learnable Decoding | M24–M48 | Develop a PAC-learning framework for quantum decoder classes. |
| WP4 | FPGA Implementation and Live Quantum Processor Deployment | M36–M60 | Implement best-performing ML-decoder in FPGA firmware. |

### WP1: Neural Decoder Architecture and Training Pipeline

Design, implement, and optimize transformer/GNN hybrid decoder architectures for surface and color codes. Establish training pipelines for both supervised and reinforcement learning objectives. Conduct hardware-aware architecture search targeting sub-microsecond inference latency. Deliverables: (i) open-source decoder library integrated with DecoderBench; (ii) at least two peer-reviewed publications on architecture and training methodology; (iii) benchmark results against MWPM and Union-Find on simulated noise models across code distances 3–9.

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**T1.1: Hybrid transformer/GNN architecture design** — Design and implement the transformer/GNN hybrid architecture with stabilizer check graph as inductive bias. Benchmark against existing neural decoder architectures on standard phenomenological noise models (M1–9).

**T1.2: Reinforcement learning training for logical error rate optimization** — Implement RL training loop with logical error rate reward. Compare RL-trained decoders against supervised-trained baselines on correlated noise models. Identify training regimes where RL provides consistent advantage (M6–18).

**T1.3: Hardware-aware architecture search and compression** — Apply structured pruning, quantization-aware training, and hardware-aware neural architecture search to push inference latency toward one microsecond. Characterize accuracy-latency Pareto frontier (M12–24).

### WP2: Hardware Noise Modeling and Experimental Validation

Extract device-specific noise profiles from IBM Quantum and partner hardware. Retrain and fine-tune ML-decoders on hardware syndrome data. Perform direct comparison of ML-decoder logical error rates against MWPM on hardware-realistic noise. Deliverables: (i) hardware-validated noise model library; (ii) peer-reviewed publication demonstrating ML-decoder advantage on realistic noise; (iii) transfer learning protocol for rapid decoder adaptation to new devices.

**T2.1: Device noise model extraction from public calibration data** — Construct high-fidelity noise models from IBM Quantum public calibration data including correlated error characterization via randomized benchmarking extensions (M12–20).

**T2.2: Hardware partnership and raw syndrome data collection** — Establish research partnership with IBM Quantum or Google Quantum AI. Collect raw syndrome measurement data from repeated stabilizer cycles on current processors (M18–32).

**T2.3: Transfer learning and hardware-adaptive decoder validation** — Fine-tune simulated-noise-trained decoders on hardware syndrome data. Conduct head-to-head comparison of ML-decoder and MWPM logical error rates on hardware-realistic noise models (M24–36).

### WP3: Theoretical Foundations of Learnable Decoding

Develop a PAC-learning framework for quantum decoder classes. Establish sample complexity bounds, VC-dimension estimates, and connections to quantum channel capacity. Investigate computational complexity of optimal decoding under correlated noise. Deliverables: (i) theoretical manuscript on PAC-learning bounds for decoder classes; (ii) hardness results for optimal decoding under correlated noise; (iii) framework connecting decoder representational capacity to information-theoretic limits.

**T3.1: PAC-learning bounds for neural decoder classes** — Define hypothesis classes of parameterized neural decoders. Establish VC-dimension and Rademacher complexity bounds. Derive sample complexity bounds for near-optimal decoding (M24–36).

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**T3.2: Connections to quantum channel capacity and information-theoretic limits** — Formalize the relationship between the generalization error of a learned decoder and the gap from the information-theoretic decoding threshold. Identify code families and noise regimes where learnable near-optimal decoders provably exist (M30–42).

**T3.3: Computational complexity of correlated-noise optimal decoding** — Establish hardness results (NP-hardness or hardness of approximation) for optimal decoding under general correlated noise models, formally motivating the heuristic ML approach (M36–48).

### WP4: FPGA Implementation and Live Quantum Processor Deployment

Implement best-performing ML-decoder in FPGA firmware. Integrate with classical control stack of partner quantum processor. Demonstrate closed-loop error correction using learned decoding policy on a live superconducting quantum processor. Deliverables: (i) FPGA-implemented ML-decoder meeting latency target; (ii) peer-reviewed publication on closed-loop ML-decoded error correction; (iii) open-source FPGA decoder firmware.

**T4.1: Fixed-point FPGA implementation of compressed neural decoder** — Implement quantized, pruned decoder network in fixed-point arithmetic on Xilinx Ultrascale+ FPGA. Optimize pipelined inference to achieve target latency for distance-5 surface code (M36–48).

**T4.2: Control stack integration and latency verification** — Integrate FPGA decoder with the classical control electronics of the partner quantum processor. Verify end-to-end latency under realistic operating conditions (M42–54).

**T4.3: Closed-loop error correction demonstration** — Demonstrate closed-loop error correction on live superconducting quantum processor using FPGA-implemented ML-decoder. Compare achieved logical error rate against MWPM baseline under identical hardware conditions (M48–60).

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## Risk Assessment and Mitigating Measures

TQML is an ambitious project operating at the frontier of three rapidly evolving fields. The following risks have been identified with concrete mitigation strategies.

Risk 1 — Real-time latency constraint (HIGH probability, HIGH impact). The one-microsecond inference target for a distance-5 surface code may not be achievable with standard neural network architectures on current FPGA hardware. Mitigation: (i) Architecture design in WP1 is explicitly latency-driven, with pruning and quantization integrated from the outset rather than applied post-hoc; (ii) the project includes a formally designed fallback to a hybrid NN+MWPM architecture in which the neural component provides correlated-error probability estimates that are fed into an MWPM solver — this hybrid approach is independently novel and publishable, and preliminary analysis suggests it can meet latency targets even if a fully neural approach cannot; (iii) the theoretical and hardware validation objectives (WP2, WP3) are independent of the FPGA latency target and will produce full scientific contributions regardless of WP4 outcomes.

Risk 2 — Hardware partner access delays (MEDIUM probability, HIGH impact). Delays in establishing a formal research partnership with IBM Quantum or Google Quantum AI could delay WP2 and WP4. Mitigation: (i) IBM Quantum's publicly accessible cloud platform and published calibration data provide an immediate and sufficient starting point for WP2 hardware noise modeling from M12 without requiring a formal partnership; (ii) the PI has existing contacts at both IBM Research Zurich and Google Quantum AI through the QIP community and ETH Zurich's institutional relationships; (iii) if no formal partnership is established by M24, the project will proceed with cloud-accessible hardware and publicly available calibration data, with WP4 scope adjusted from closed-loop demonstration to validated open-loop syndrome prediction — which remains a highly significant and publishable result.

Risk 3 — Theoretical results limited to specific code families (MEDIUM probability, MEDIUM impact). PAC-learning bounds established for surface codes may not generalize to other topological code families. Mitigation: (i) The theoretical programme is explicitly designed to cover both surface codes and color codes in parallel, maximizing scope of generalizability from the outset; (ii) negative results — establishing that certain code families do not admit efficiently learnable decoders — are themselves significant theoretical contributions and publishable outcomes; (iii) the theoretical work in WP3 is self-contained and does not depend on the empirical results of WP1 or WP2.

Risk 4 — RL training instability (LOW probability, MEDIUM impact). Reinforcement learning for decoder optimization may suffer from reward sparsity and training instability for large code distances. Mitigation: Curriculum learning (training on small distances first, then progressively scaling) and shaped reward functions (intermediate rewards for local correction quality) will be implemented from the start of T1.2. These approaches have established effectiveness in the deep RL literature and are well matched to the hierarchical structure of syndrome decoding.

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## Team and Resources

The TQML team will consist of Dr. Elena Vasquez (PI, minimum 50% time commitment) supported by two postdoctoral researchers and two PhD students recruited in M1–M3.

Postdoctoral Researcher 1 (experimental/hardware, M1–60): A researcher with expertise in superconducting quantum hardware and FPGA-based control systems. This individual will lead the experimental activities of WP2 and WP4, managing hardware partnership interactions, noise model extraction, and FPGA implementation. The combination of quantum hardware and digital systems engineering expertise in a single researcher is rare but essential for the hardware-to-software integration that TQML requires. Postdoctoral Researcher 2 (theoretical, M1–60): A researcher with expertise in quantum information theory and computational learning theory. This individual will lead WP3, driving the PAC-learning framework and complexity-theoretic investigations. PhD Student 1 (machine

learning / neural decoder design, M1–60): Will focus on the architecture design and training pipeline in WP1, working closely with the PI. PhD Student 2 (theory and numerical simulation, M1–60): Will support WP3 theoretical work and conduct large-scale numerical simulations of decoder performance as a function of code distance, noise model, and architecture complexity.

ETH Zurich provides world-class research infrastructure including access to GPU computing clusters through ETH Scientific IT Services (Euler cluster), high-performance networking for distributed training, and electronics facilities relevant to FPGA development. The host institution's established relationship with IBM Research Zurich (located approximately 15 km away) and access to the IBM Quantum Network through the ETH quantum computing initiative provide direct institutional support for the hardware partnership pursued in WP2 and WP4.

## Budget Justification

The total requested budget of EUR 1,490,000 over 60 months is justified as follows.

Personnel (EUR 1,000,000, 67%): Two postdoctoral researchers at EUR 300,000 each over 60 months (EUR 600,000 total), reflecting Swiss postdoctoral salary levels including ETH Zurich employer contributions. Two PhD students at EUR 200,000 each over 60 months (EUR 400,000 total), reflecting ETH Zurich PhD stipend levels and associated costs. These personnel costs are essential: the four scientific streams of TQML require simultaneous expertise in neural architecture design, superconducting hardware, quantum information theory, and computational learning theory. No single researcher can cover all four areas, and the project cannot be descoped without losing essential scientific coverage.

Computing resources (EUR 200,000, 13%): Large-scale GPU cluster access is essential for the neural architecture search, RL training at scale, and Monte Carlo simulation of syndrome data for large code distances (d = 7 to d = 11). Training a single decoder model for a distance-7 surface code under correlated noise requires approximately 10^8 training samples and several hundred GPU-hours. This budget covers cloud GPU compute (estimated 50,000 GPU-hours over the project lifetime) for phases where local ETH Euler cluster capacity is insufficient, and licenses for hardware-aware architecture search tools.

FPGA development hardware (EUR 120,000, 8%): Two Xilinx Ultrascale+ development boards and associated evaluation kits (EUR 60,000), specialized high-speed data acquisition hardware for interfacing with quantum control electronics (EUR 40,000), and test and measurement equipment for latency verification (EUR 20,000).

Travel and conferences (EUR 80,000, 5%): Annual attendance by the PI and team members at QIP, APS March Meeting, and IEEE Quantum Week, as well as visits to hardware partner sites. International visibility is essential for maintaining the collaborations on which the hardware partnership in WP2 depends.

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Other costs (EUR 90,000, 6%): Open-access publication charges (Nature Physics, PRX Quantum, Physical Review Letters); consumables; software licenses; and costs associated with hardware partner visits and extended collaborative stays.

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[1] Vasquez, E. et al. (2021). Neural network decoders for small-distance surface codes. Nature Physics, 17(4), 429–437.

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# Funding ID

## Ongoing Grants

No ongoing grants held by Dr. Elena Vasquez at the time of submission. The ETH Zurich Postdoctoral Fellowship (2020–2022) has concluded. No other external funding is currently active.

## Submitted Applications

No other applications for this project or substantially overlapping projects have been submitted to any funding agency at the time of submission of this proposal.

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